

Engineering Mechanics: ME101

Lecture 2: Statics



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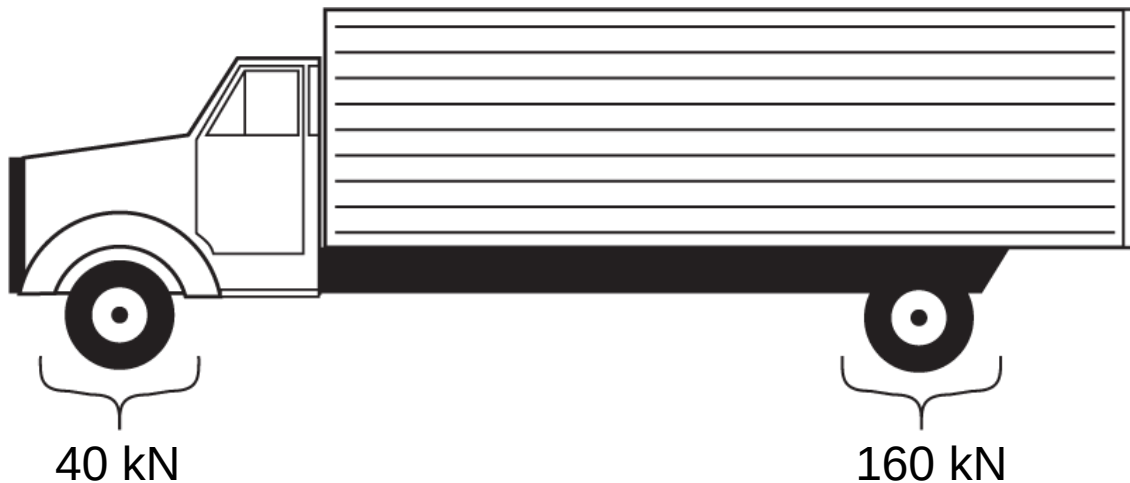
Classification of forces

- **Concentrated force:** A single force with a line of action and a point of application.
- **Volume force:** It is a force that is distributed over volume of the a body or portion.
- **Area force:** It occurs in the region where two bodies are in contact.
- **Active forces:** Physically prescribed forces in a mechanical system
- **Reactive force:** Generated if the freedom of movement of a body is constraint.

Mechanics: Force idealization

- **Concentrated Force**

- **Line of action of weight** through the **centre of gravity** of the body
- **Area over which the load is applied is very small** compared to the **overall size** of the body



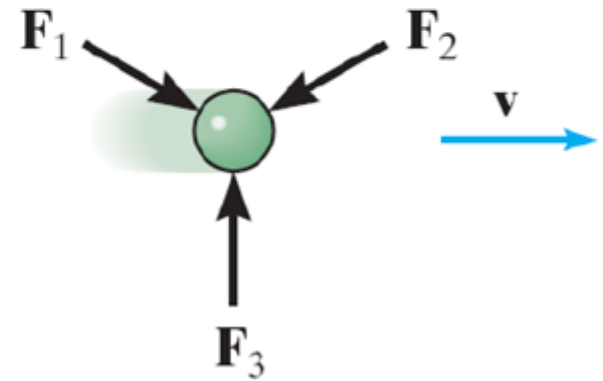
Ex: Contact Force
between a wheel
and ground.

Mechanics: Newton's Three Laws of Motion

Basis of formulation of rigid body mechanics.

First Law: A particle originally at rest, or moving in a straight line with constant velocity, tends to remain in this state provided the particle is not subjected to an unbalanced force.

First law contains the principle of the equilibrium of forces → main topic of concern in Statics



□ It provides the Principle of force equilibrium (statics)

Mechanics: Newton's Three Laws of Motion

- **Basis of rigid body dynamics**

- **Second Law:** A particle of mass “ m ” acted upon by an unbalanced force “ F ” experiences an acceleration “ a ” that has the same direction as the force and a magnitude that is directly proportional to the force

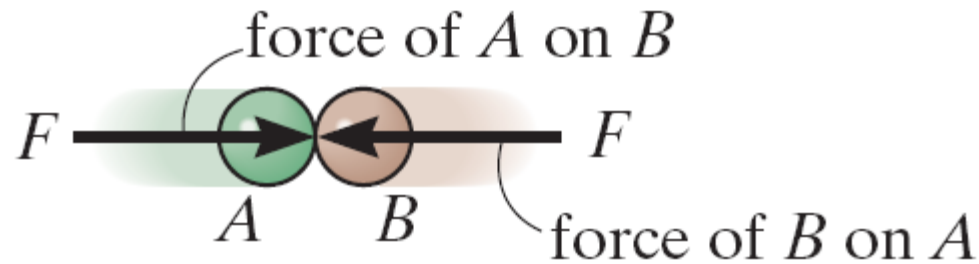


Accelerated motion

❖ It forms the basis of most of the analysis of dynamics

Mechanics: Newton's Three Laws of Motion

- Application in both statics and dynamics
 - **Third Law**: The mutual forces of action and reaction between two particles are equal, opposite and collinear.



Action – reaction

- Third law is basic to our understanding of Force → Forces always occur in pairs of equal and opposite forces.

Mechanics: Scalars and Vectors

- Scalar
 - Only **magnitude** is associated with it
 - *e.g.*, time, volume, density, speed, energy, mass etc.
- Vector
 - Possess **direction** as well as **magnitude**
 - Parallelogram law of addition (and the triangle law)
 - *e.g.*, displacement, velocity, acceleration etc.
- Tensor (it depends on plane of orientation also)
 - *e.g.*, stress (3×3 components)

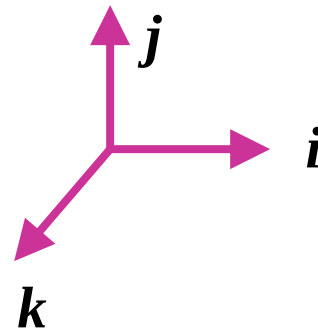
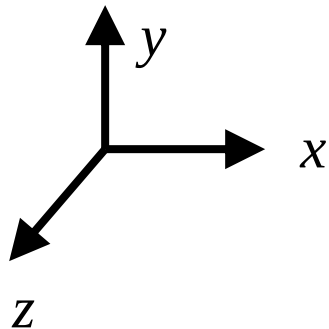
Mechanics: Scalars and Vectors

- A Vector **V** can be written as: $\mathbf{V} = V\mathbf{n}$

V = magnitude of **V**

$\mathbf{n}$ = unit vector whose magnitude is one and whose direction coincides with that of **V**

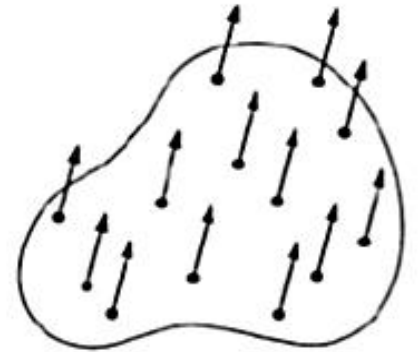
- Unit vector can be formed by dividing any vector, such as the geometric position vector, by its length or magnitude
- Vectors represented by Bold and Non-Italic letters (**V**)
- Magnitude of vectors represented by Non-Bold, Italic letters (V)



Type of Vectors

Free Vector: whose action is not confined to or associated with a unique line in space

Ex: Movement of a body without rotation.

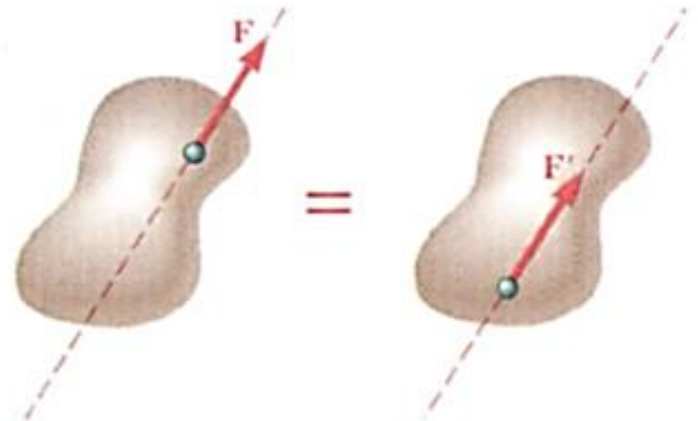


Sliding Vector: has a unique line of action in space but not a unique point of application

Ex: External force on a rigid body

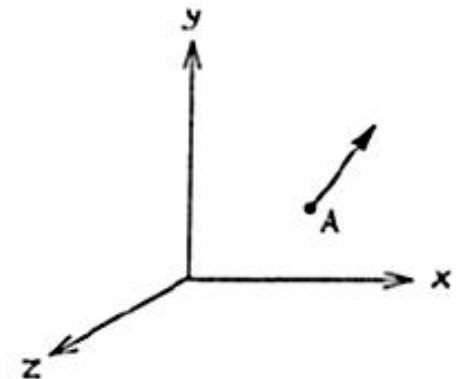
→ Principle of Transmissibility

→ Imp in Rigid Body Mechanics



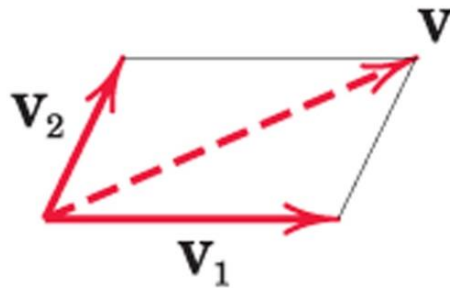
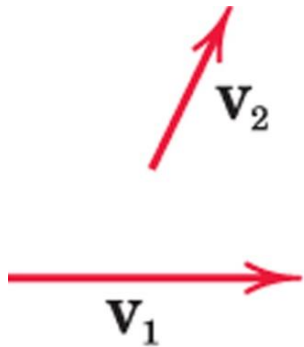
Fixed Vector: for which a unique point of application is specified

Ex: Action of a force on deformable body

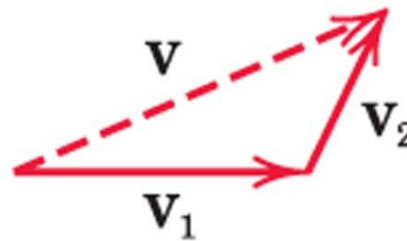


Laws of vectors addition

– Equivalent vector $\mathbf{v} = \mathbf{v}_1 + \mathbf{v}_2$ (Vector Sum)

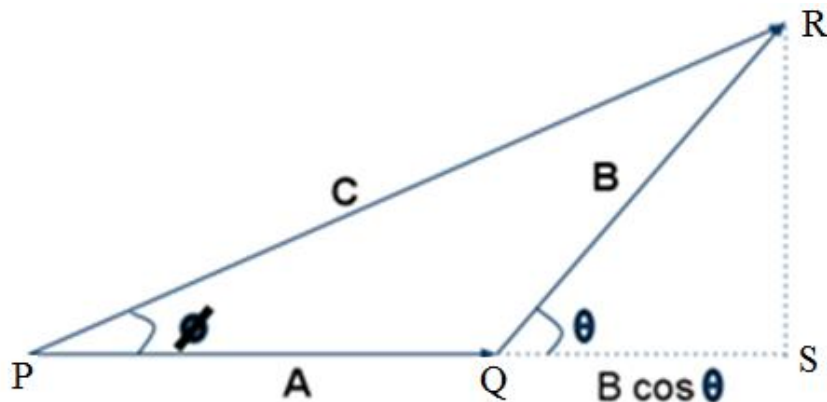


✓ Parallelogram law of vector addition



✓ Triangle law of vector addition

❖ Triangle law of vector addition:



' θ ' is the angle between $\vec{A}$ & $\vec{B}$.

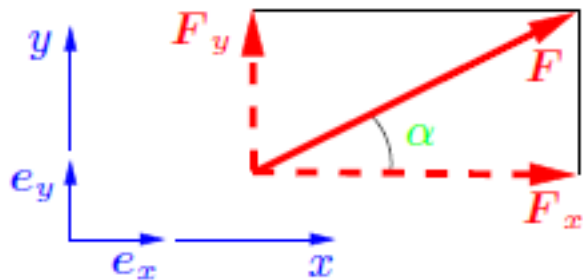
' ϕ ' is the angle between $\vec{C}$ & $\vec{A}$.

$$|\vec{C}| = \sqrt{A^2 + B^2 + 2AB \cos \theta}$$

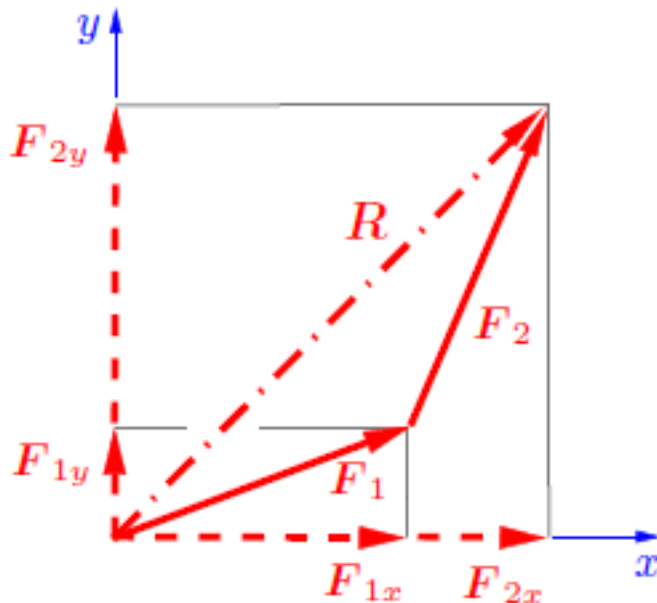
$$\phi = \tan^{-1} \left(\frac{B \sin \theta}{A + B \cos \theta} \right)$$

$\vec{C}$ makes an angle ϕ with the vector $\vec{A}$.

Addition of Forces in a Plane



$$F_x = F \cos \alpha, \quad F_y = F \sin \alpha,$$
$$F = \sqrt{F_x^2 + F_y^2}, \quad \tan \alpha = \frac{F_y}{F_x}.$$



$$R_x = F_{1x} + F_{2x}, \quad R_y = F_{1y} + F_{2y}$$

$$R_x = \sum F_{ix}, \quad R_y = \sum F_{iy}$$

$$R = \sqrt{R_x^2 + R_y^2}, \quad \tan \alpha_R = \frac{R_y}{R_x}.$$

Resolution of vectors in Cartesian coordinate

□ Resultant Force vector

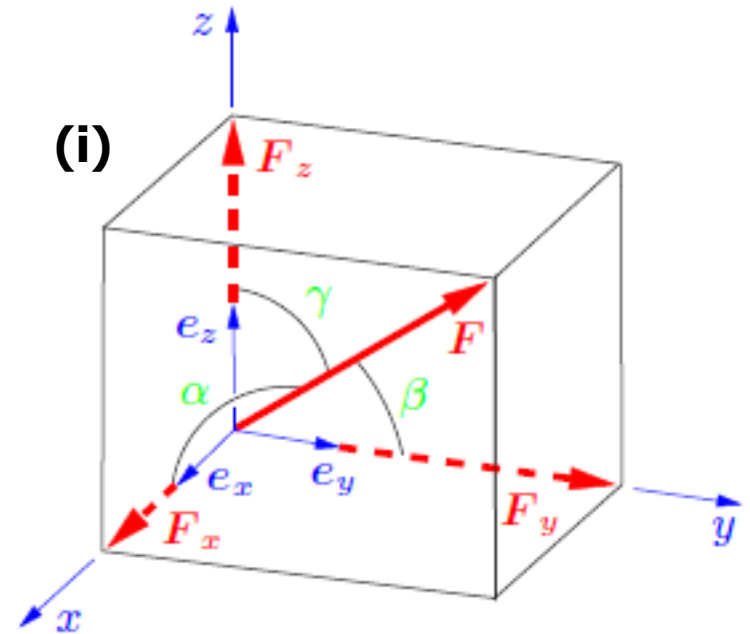
$$\mathbf{F} = F_x + F_y + F_z = F_x \mathbf{e}_x + F_y \mathbf{e}_y + F_z \mathbf{e}_z \quad (\text{i})$$

where $\mathbf{e}$ = unit vector

unit vectors $\mathbf{e}_x$, $\mathbf{e}_y$, $\mathbf{e}_z$

□ Force vector magnitude

$$F = \sqrt{F_x^2 + F_y^2 + F_z^2} \quad (\text{ii})$$



✓ The direction angles and therefore the direction of the force follow from:

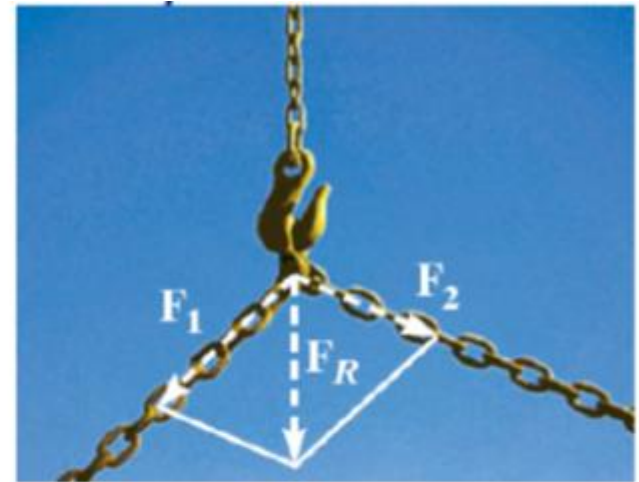
$$\cos \alpha = \frac{F_x}{F}, \quad \cos \beta = \frac{F_y}{F}, \quad \cos \gamma = \frac{F_z}{F} \quad (\text{iii})$$

Procedure for Analysis

Parallelogram Law (Graphical)

Resultant Force (diagonal)

Components (sides of parallelogram)



Algebraic Solution

Using the coordinate system

Cosine law:

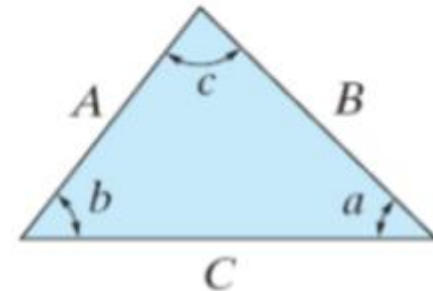
$$C = \sqrt{A^2 + B^2 - 2AB \cos c}$$

Sine law:

$$\frac{A}{\sin a} = \frac{B}{\sin b} = \frac{C}{\sin c}$$

Trigonometry (Geometry)

Resultant Force and Components
from Law of Cosines and Law of Sines



Force System

- ❑ A systems of concentrated forces that have a common point of application are investigated. Such forces are **called concurrent forces**.
- ❑ If all the forces acting on a body act in a plane, they are called **coplanar forces**.

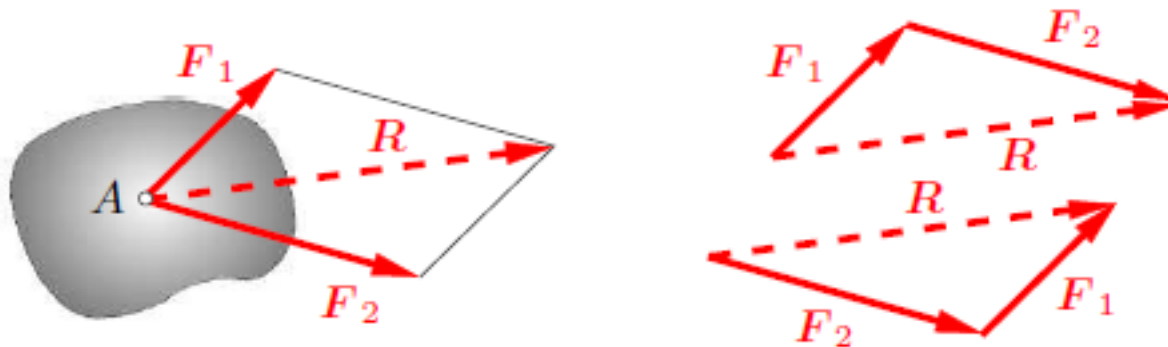


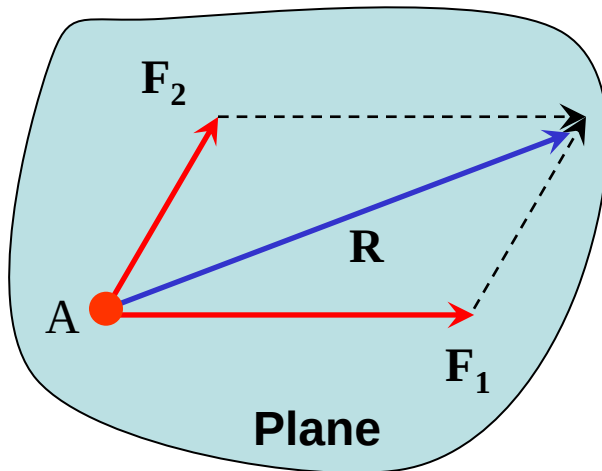
Fig. Addition of Forces in a Plane

- ❖ Since in this case the force vectors are sliding vectors, they may be applied **at any point along their lines of action without changing their effect on the body** (principle of transmissibility)

Force Systems

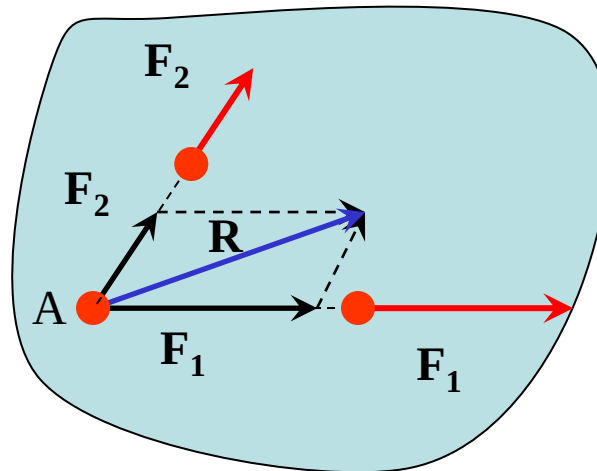
- **Concurrent forces**

- Lines of action intersect at a point



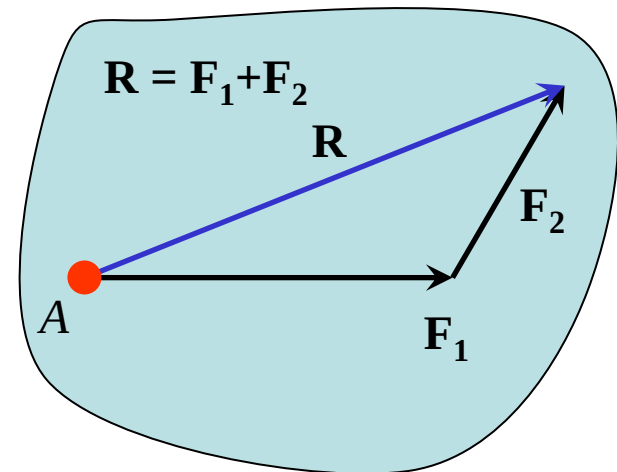
Concurrent Forces

F_1 and F_2



Principle of

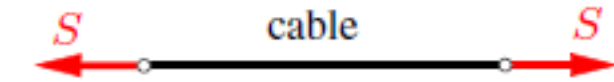
Transmissibility



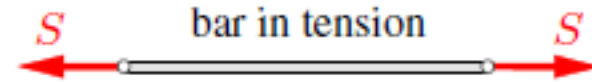
$$R = F_1 + F_2$$

❖ **Principle of transmissibility**, states that a force may be applied at any point on its given line of action without altering the resultant effects of the force to the rigid body on which it acts.

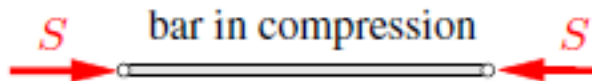
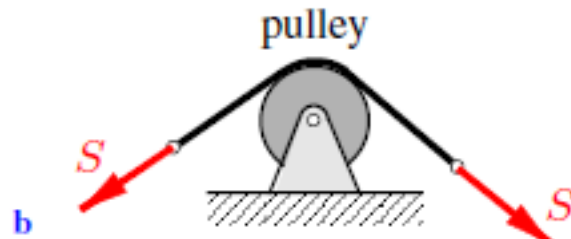
Example of coplanar forces



a



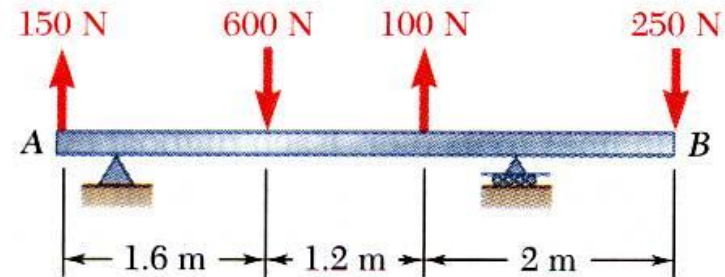
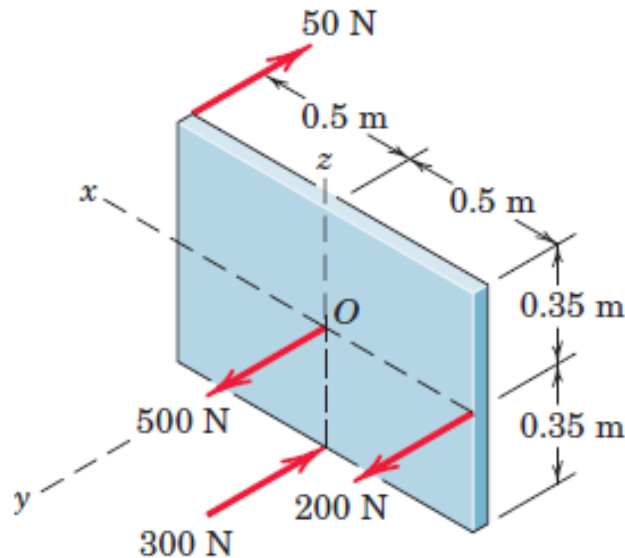
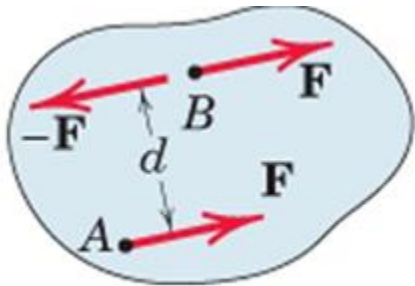
b



Force Systems

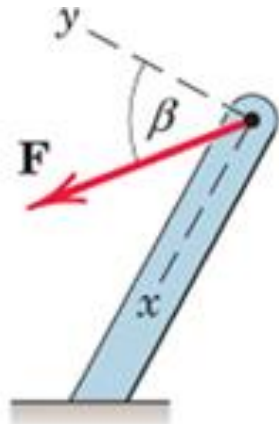
- **Parallel Forces**

- Lines of action do not intersect at a point

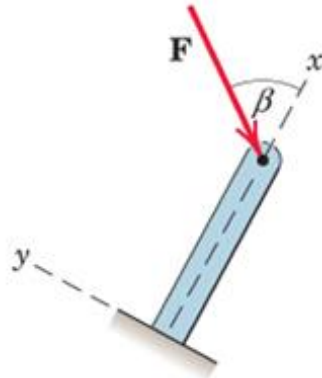


Components of a Force

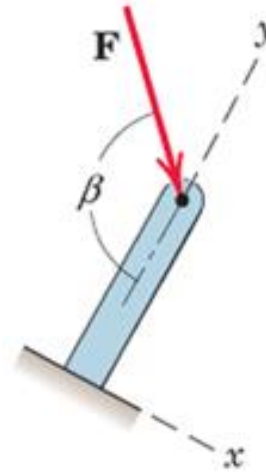
- Examples



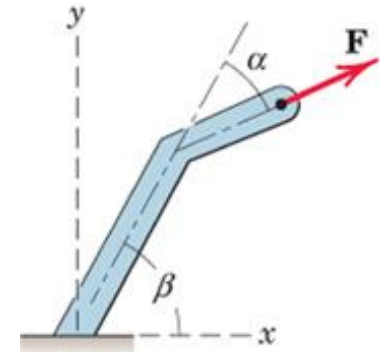
$$F_x = F \sin \beta$$
$$F_y = F \cos \beta$$



$$F_x = -F \cos \beta$$
$$F_y = -F \sin \beta$$



$$F_x = F \sin(\pi - \beta)$$
$$F_y = -F \cos(\pi - \beta)$$



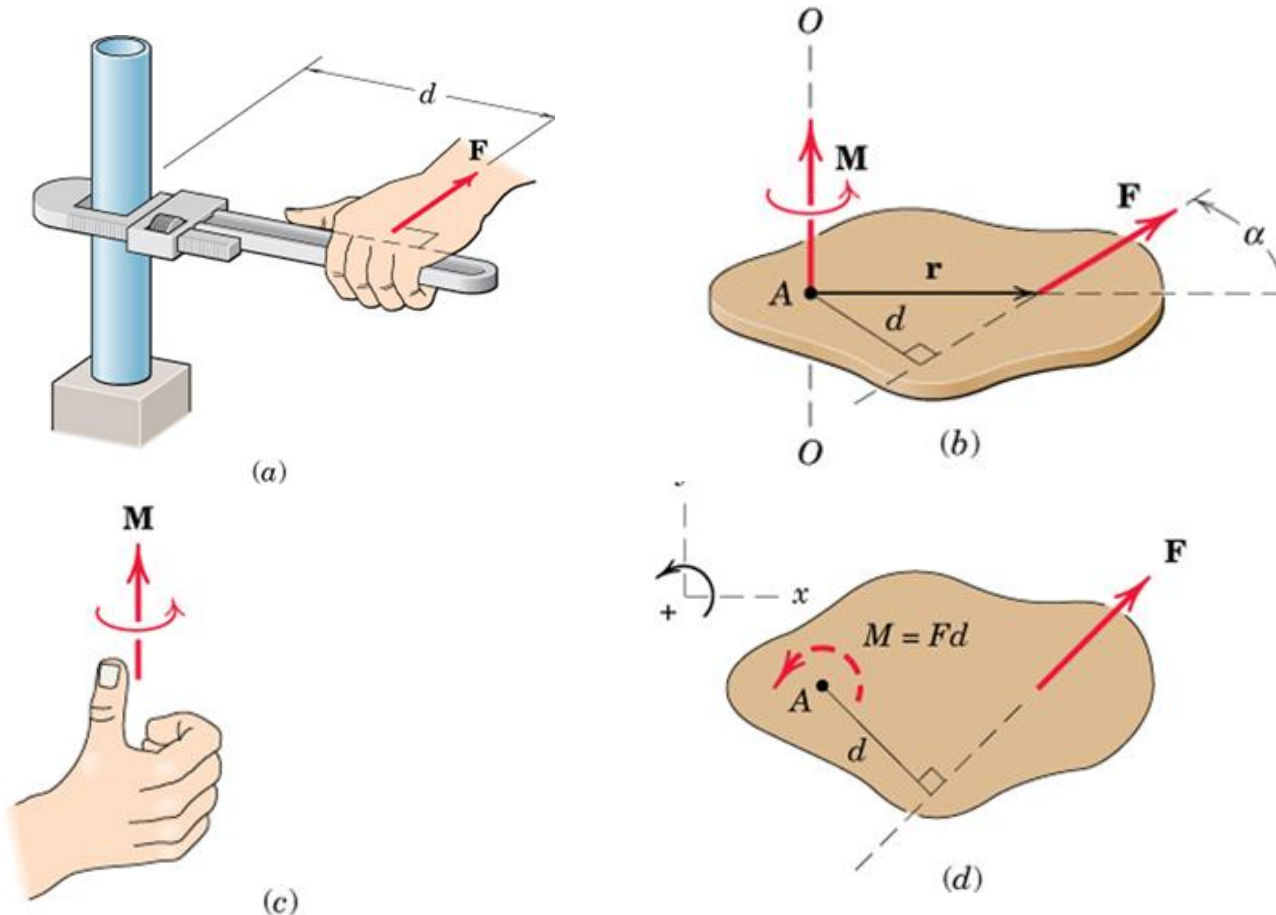
$$F_x = F \cos(\beta - \alpha)$$
$$F_y = F \sin(\beta - \alpha)$$

Moment of a Force (Torque)

- **Moment of a Force (F) @ point A**

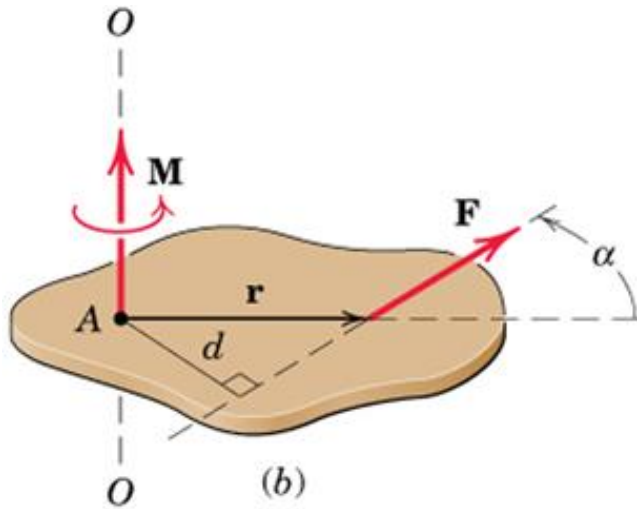
$$\mathbf{M}_O = \mathbf{r} \times \mathbf{F}$$

$\mathbf{r}$ = position vector directed from O to any point on the line of action of $\mathbf{F}$



Moment of a Force

- ❖ **F** tends to rotate the body about an axis along **M_O**



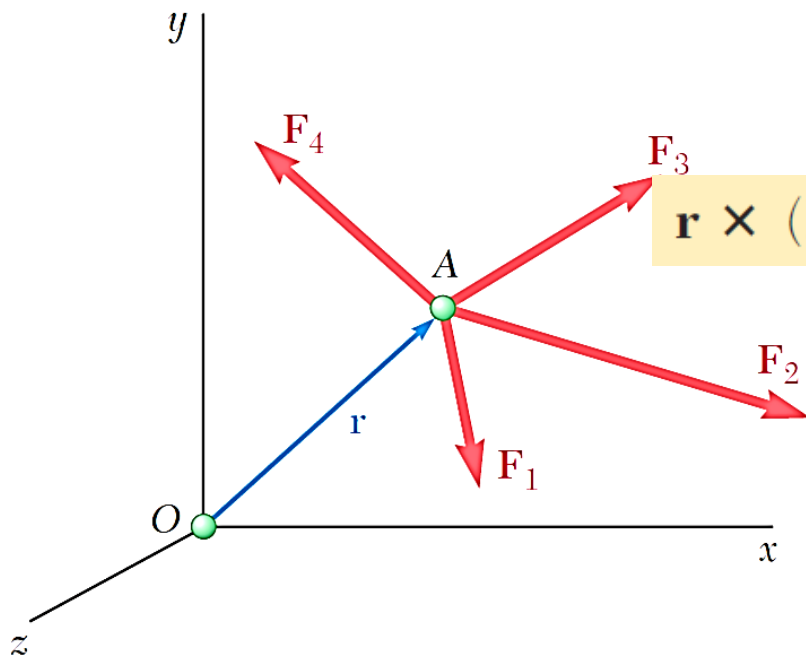
$$M_O = rF \sin \alpha = F(r \sin \alpha) = Fd$$

- ❖ Moment arm – $d = r \sin \alpha$

Moment of a System of Concurrent Forces

Varignon's Theorem

- **Moment of the resultant of a system of concurrent forces about a point is equal to the sum of the moments of the individual forces about the same point**



$$\mathbf{r} \times (\mathbf{F}_1 + \mathbf{F}_2 + \cdots) = \mathbf{r} \times \mathbf{F}_1 + \mathbf{r} \times \mathbf{F}_2 + \cdots$$

Rectangular Components of Moment

The moment of $\vec{F}$ about B ,

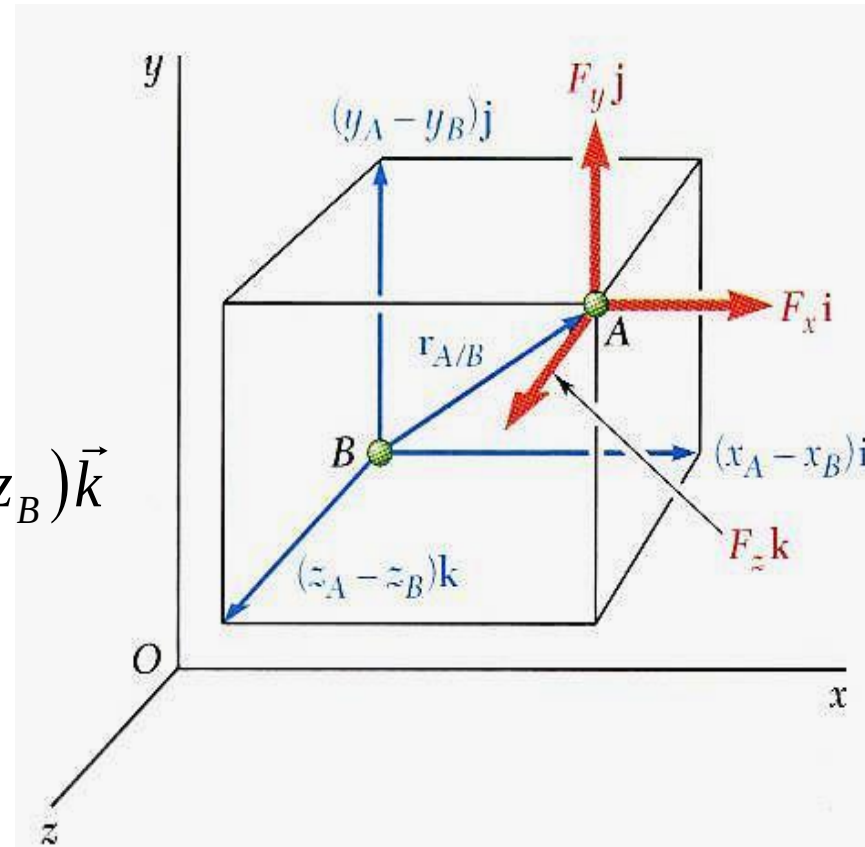
$$\vec{M}_B = \vec{r}_{A/B} \times \vec{F}$$

$$\vec{r}_{A/B} = \vec{r}_A - \vec{r}_B$$

$$= (x_A - x_B)\vec{i} + (y_A - y_B)\vec{j} + (z_A - z_B)\vec{k}$$

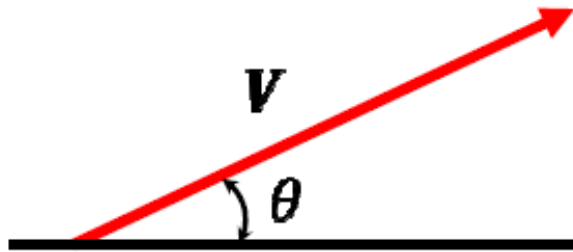
$$\vec{F} = F_x\vec{i} + F_y\vec{j} + F_z\vec{k}$$

$$\vec{M}_B = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ (x_A - x_B) & (y_A - y_B) & (z_A - z_B) \\ F_x & F_y & F_z \end{vmatrix}$$

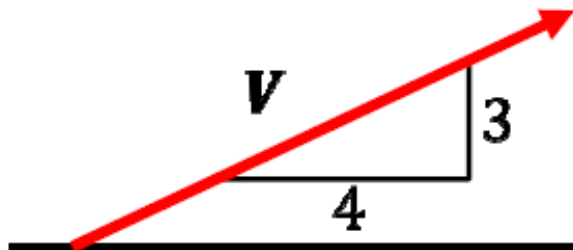


Vector Representation

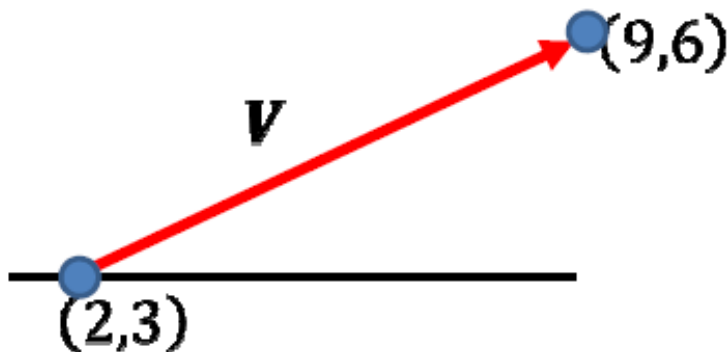
Vector



$$\mathbf{V} = V(\cos\theta\mathbf{i} + \sin\theta\mathbf{j})$$



$$\mathbf{V} = V \left(\frac{4\mathbf{i} + 3\mathbf{j}}{\sqrt{4^2 + 3^2}} \right)$$

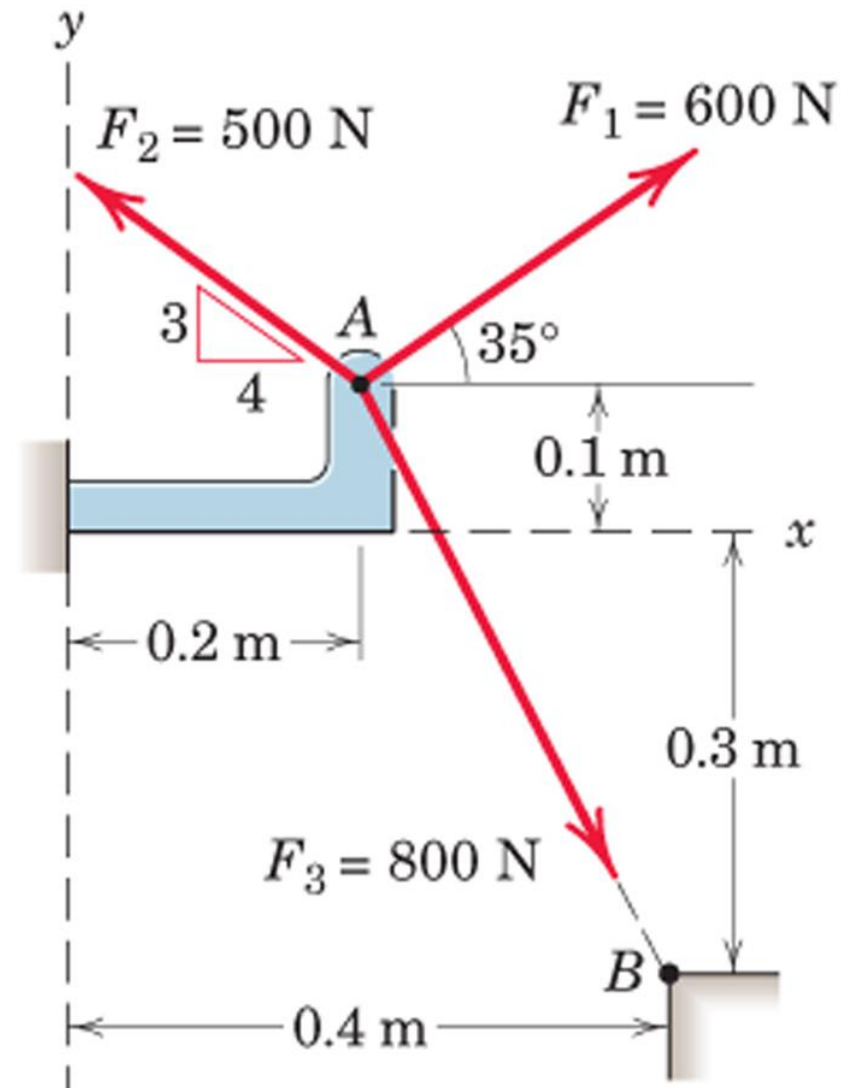


$$\mathbf{V} = V \left(\frac{(9-2)\mathbf{i} + (6-3)\mathbf{j}}{\sqrt{(9-2)^2 + (6-3)^2}} \right)$$

Components of a Force

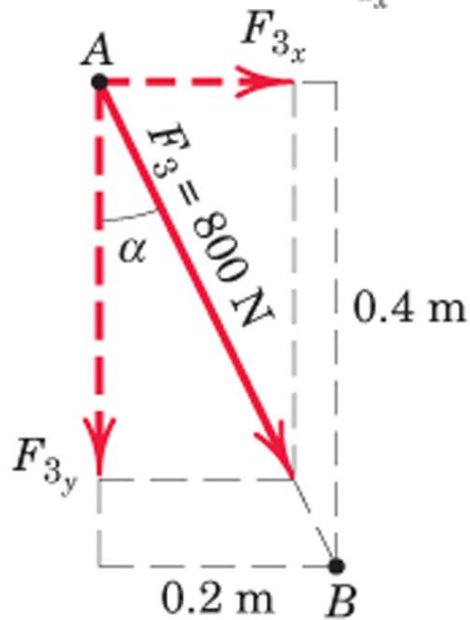
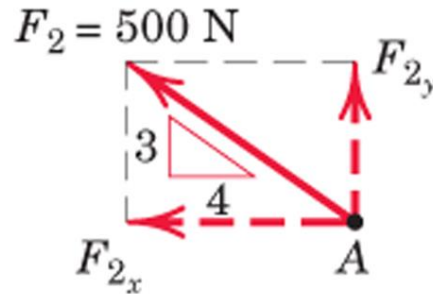
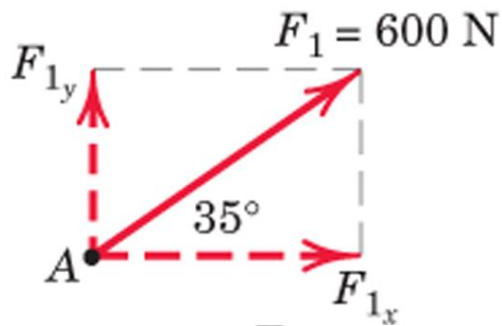
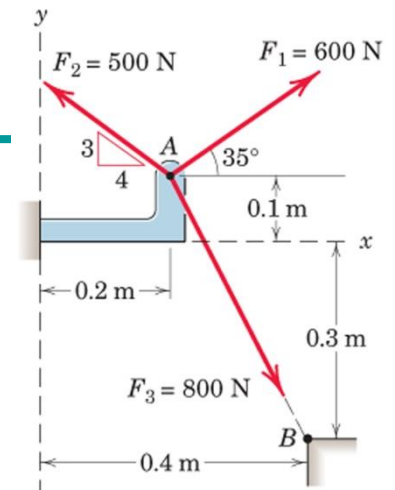
Example 1:

Determine the x and y scalar components of F_1 , F_2 , and F_3 acting at point A of the bracket



Components of Force

Solution:



$$F_{1x} = 600 \cos 35^\circ = 491 \text{ N}$$

$$F_{1y} = 600 \sin 35^\circ = 344 \text{ N}$$

$$F_{2x} = -500\left(\frac{4}{5}\right) = -400 \text{ N}$$

$$F_{2y} = 500\left(\frac{3}{5}\right) = 300 \text{ N}$$

$$\alpha = \tan^{-1} \left[\frac{0.2}{0.4} \right] = 26.6^\circ$$

$$F_{3x} = F_3 \sin \alpha = 800 \sin 26.6^\circ = 358 \text{ N}$$

$$F_{3y} = -F_3 \cos \alpha = -800 \cos 26.6^\circ = -716 \text{ N}$$

Components of Force

Vector approach: Scalar components of $\mathbf{F}_3$ can be obtained by writing F_3 as a magnitude times a unit vector $\mathbf{n}_{AB}$ in the direction of the line segment AB.

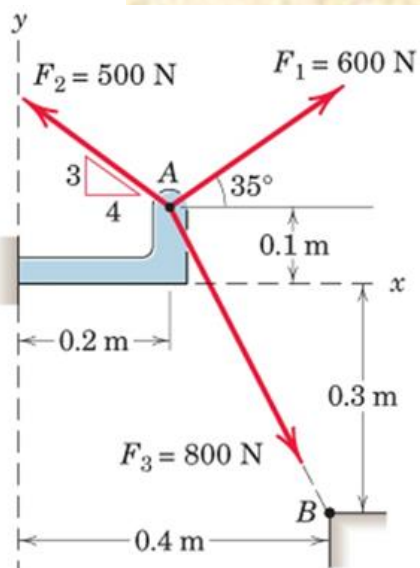
$$\mathbf{F}_3 = F_3 \mathbf{n}_{AB} = F_3 \frac{\overrightarrow{AB}}{AB} = 800 \left[\frac{0.2\mathbf{i} - 0.4\mathbf{j}}{\sqrt{(0.2)^2 + (-0.4)^2}} \right]$$

$$= 800[0.447\mathbf{i} - 0.894\mathbf{j}]$$

$$= 358\mathbf{i} - 716\mathbf{j} \text{ N}$$

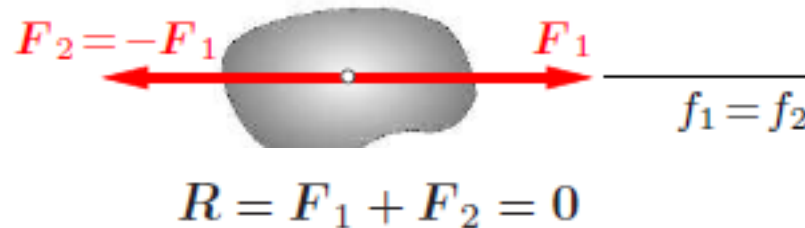
$$F_{3x} = 358 \text{ N}$$

$$F_{3y} = -716 \text{ N}$$

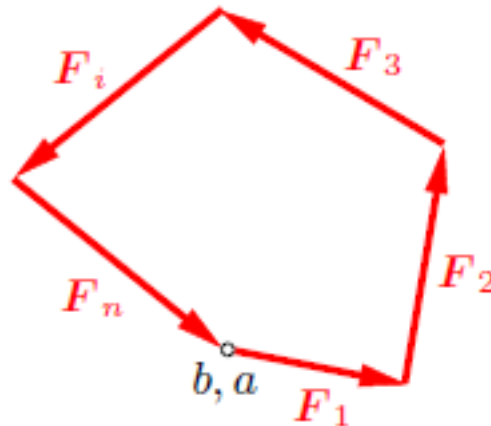


Equation of Equilibrium

- ❑ **Two forces are in equilibrium** if they are oppositely directed on the same line of action and have the same magnitude.



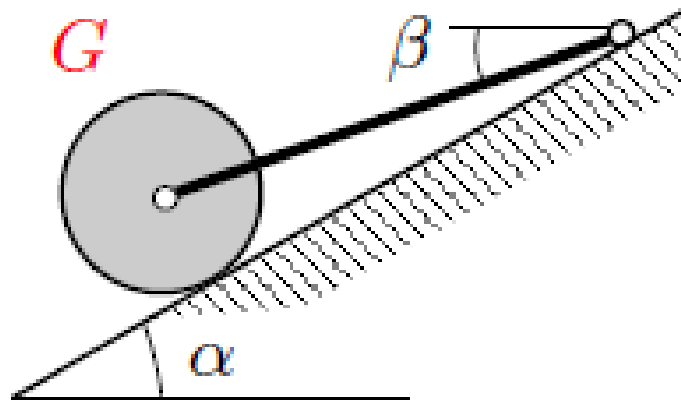
- ❖ **A system of concurrent forces is in equilibrium** if the resultant is zero.



$$R = \sum F_i = 0$$

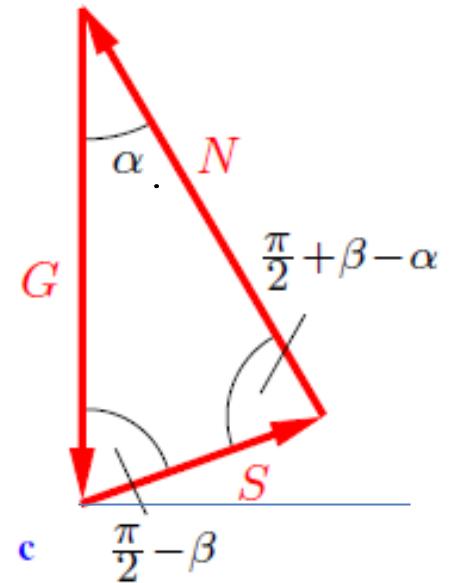
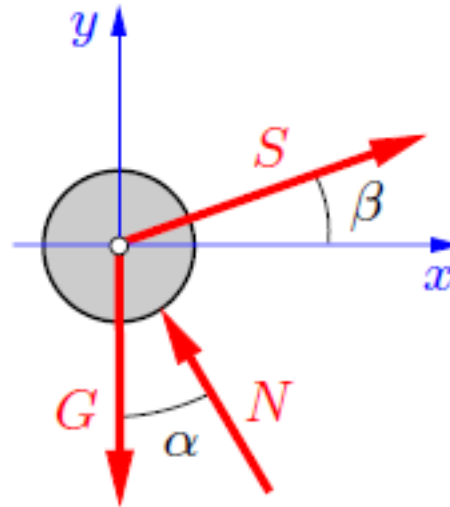
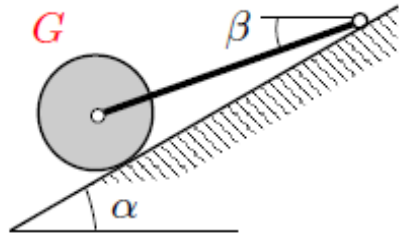
Problem

- **Ex. 4:** A wheel with weight G is held on a smooth inclined plane by a cable (fig). Determine the force in the cable and the contact force between the plane and the wheel.



Solution

□ Steps



❖ Based on Lamis theorem

➤ Graphical

$$\underline{\underline{S}} = G \frac{\sin \alpha}{\sin(\frac{\pi}{2} + \beta - \alpha)} = \underline{\underline{G \frac{\sin \alpha}{\cos(\alpha - \beta)}}},$$

$$\underline{\underline{N}} = G \frac{\sin(\frac{\pi}{2} - \beta)}{\sin(\frac{\pi}{2} + \beta - \alpha)} = \underline{\underline{G \frac{\cos \beta}{\cos(\alpha - \beta)}}}.$$

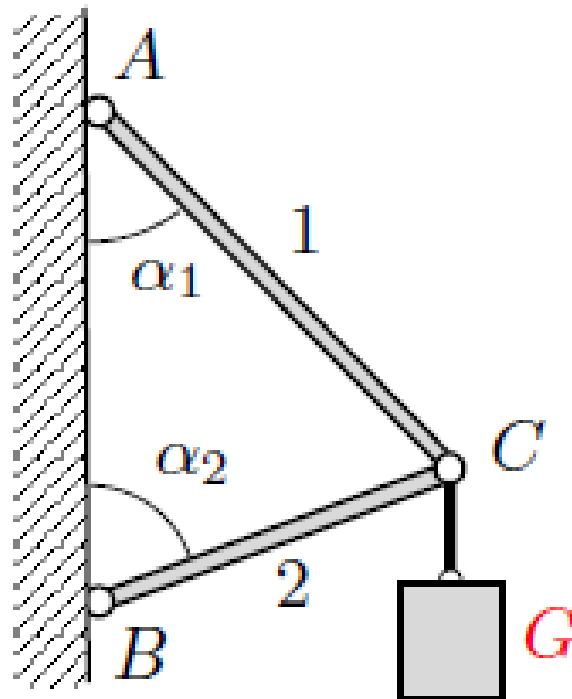
❖ Equation of equilibrium

$$\sum F_{ix} = 0 : \quad S \cos \beta - N \sin \alpha = 0 ,$$

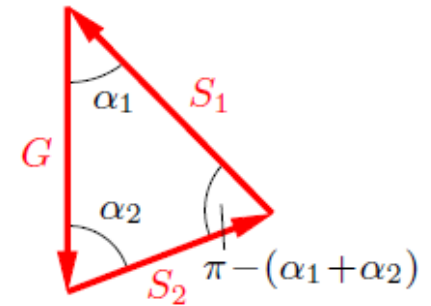
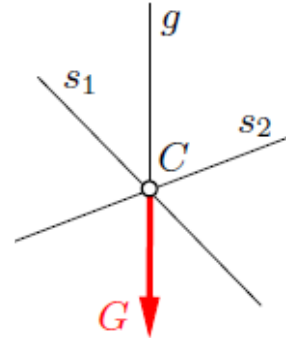
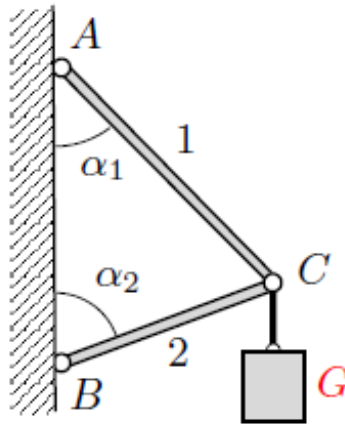
$$\sum F_{iy} = 0 : \quad S \sin \beta + N \cos \alpha - G = 0 .$$

Problem

- **Ex.5:** Two bars 1 and 2 are attached at A and B to a wall by smooth pins. They are pin-connected at C and subjected to a weight G. Calculate the forces in the bars



Solution



❖ Based on Lamis theorem

$$\underline{\underline{S_1 = G \frac{\sin \alpha_2}{\sin(\alpha_1 + \alpha_2)}}, \quad \underline{\underline{S_2 = G \frac{\sin \alpha_1}{\sin(\alpha_1 + \alpha_2)}}.}$$

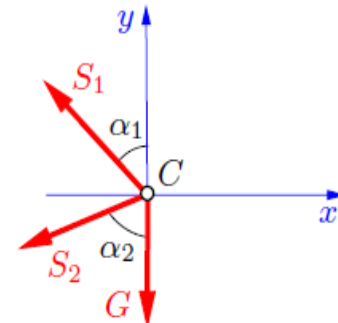
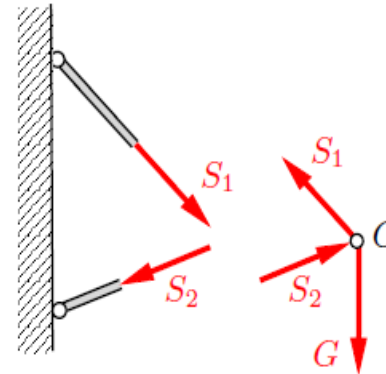
❖ Equation of equilibrium

$$\rightarrow : -S_1 \sin \alpha_1 - S_2 \sin \alpha_2 = 0,$$

$$\uparrow : S_1 \cos \alpha_1 - S_2 \cos \alpha_2 - G = 0$$

lead to

$$S_1 = G \frac{\sin \alpha_2}{\sin(\alpha_1 + \alpha_2)}, \quad S_2 = -G \frac{\sin \alpha_1}{\sin(\alpha_1 + \alpha_2)}.$$



Three-dimensional Force System

□ A force can be resolved uniquely into three components in space

$$\mathbf{F} = \mathbf{F}_x + \mathbf{F}_y + \mathbf{F}_z = F_x \mathbf{e}_x + F_y \mathbf{e}_y + F_z \mathbf{e}_z$$

✓ The magnitude and the direction of $\mathbf{F}$ are given by

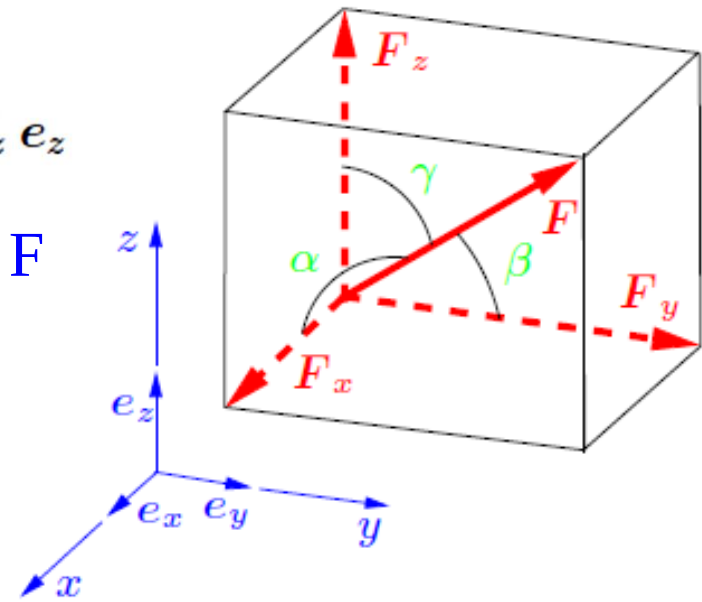
$$F = \sqrt{F_x^2 + F_y^2 + F_z^2}$$

$$\cos \alpha = \frac{F_x}{F}, \quad \cos \beta = \frac{F_y}{F}, \quad \cos \gamma = \frac{F_z}{F}$$

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

✓ For multiple forces

$$R_x = \sum F_{ix}, \quad R_y = \sum F_{iy}, \quad R_z = \sum F_{iz}$$



Three-dimensional Force System

□ The magnitude and direction of $\mathbf{R}$ follow as

$$R = \sqrt{R_x^2 + R_y^2 + R_z^2},$$
$$\cos \alpha_R = \frac{R_x}{R}, \quad \cos \beta_R = \frac{R_y}{R}, \quad \cos \gamma_R = \frac{R_z}{R}.$$

□ A spatial system of concurrent forces is in equilibrium if the resultant is the zero vector

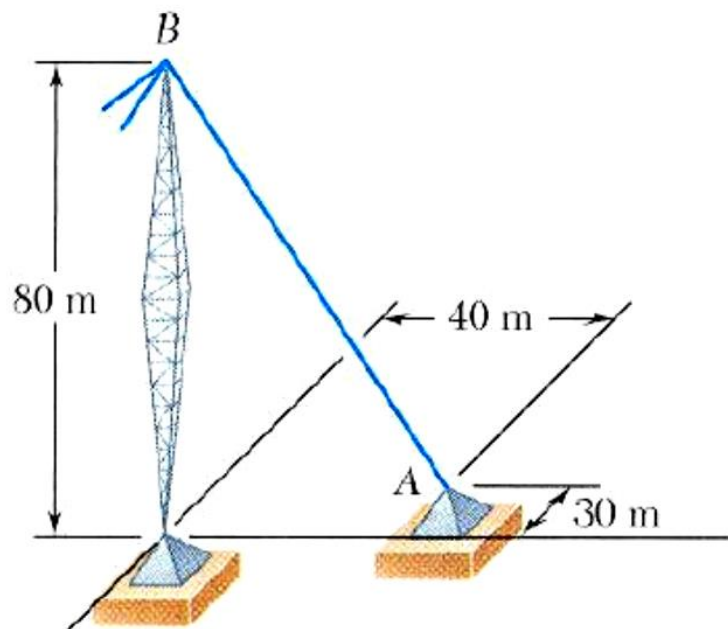
$$\mathbf{R} = \sum \mathbf{F}_i = \mathbf{0}.$$

$$\sum F_{ix} = 0, \quad \sum F_{iy} = 0, \quad \sum F_{iz} = 0,$$

Rectangular Components in Space

Example: The tension in the guy wire is 2500 N. Determine:

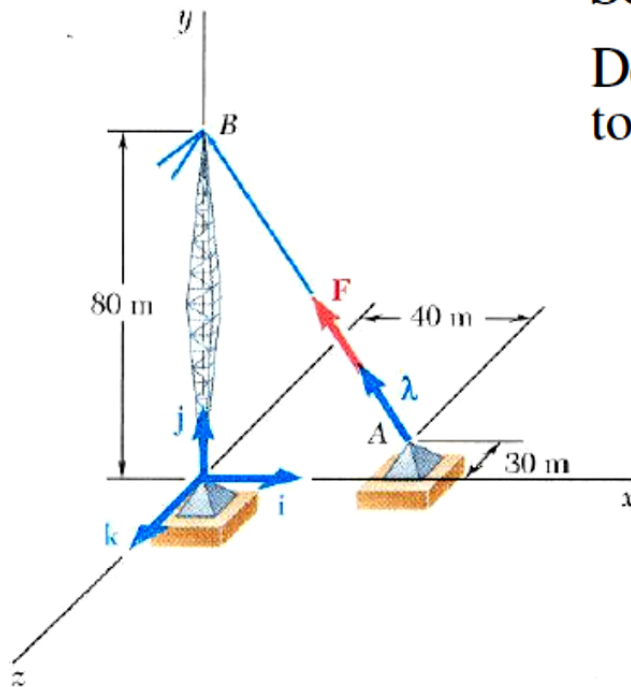
- components F_x F_y F_z of the force acting on the bolt at A,
- the angles q_x q_y q_z defining the direction of the force



SOLUTION:

- Based on the relative locations of the points A and B, determine the unit vector pointing from A towards B.
- Apply the unit vector to determine the components of the force acting on A.
- Noting that the components of the unit vector are the direction cosines for the vector, calculate the corresponding angles.

Rectangular Components in Space



Solution

Determine the unit vector pointing from A towards B. $\mathbf{AB} = (x_B - x_A)\mathbf{i} + (y_B - y_A)\mathbf{j} + (z_B - z_A)\mathbf{k}$

$$\mathbf{AB} = -40\mathbf{i} + 80\mathbf{j} + 30\mathbf{k}$$

$$AB = \sqrt{(-40)^2 + (80)^2 + (30)^2} = 94.3$$

$$\boldsymbol{\lambda} = \frac{\mathbf{AB}}{AB} = \frac{-40\mathbf{i} + 80\mathbf{j} + 30\mathbf{k}}{94.3}$$

$$= -0.424\mathbf{i} + 0.848\mathbf{j} + 0.318\mathbf{k}$$

Determine the components of the force.

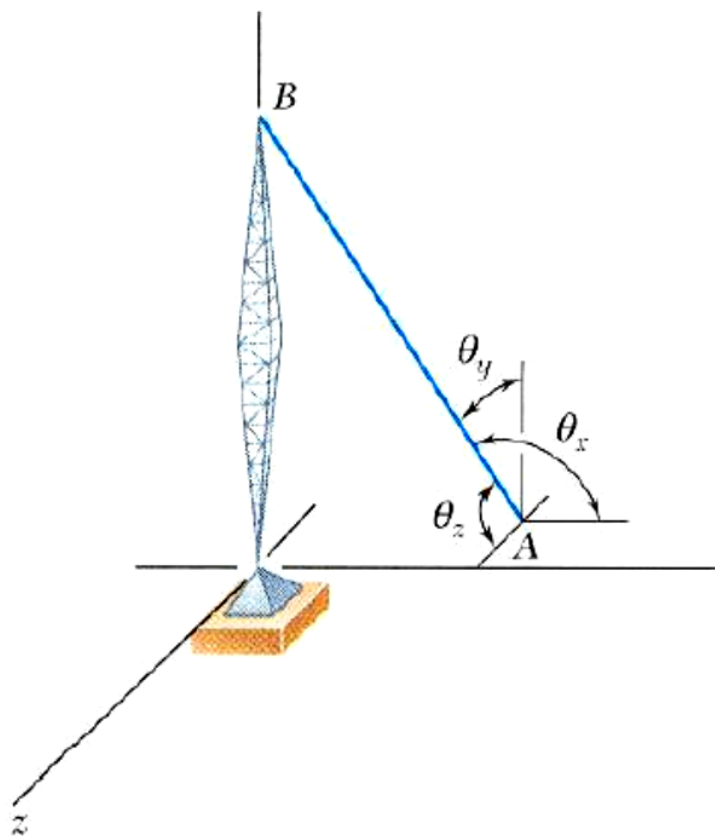
$$\begin{aligned}\mathbf{F} = F\boldsymbol{\lambda} &= 2500(-0.424\mathbf{i} + 0.848\mathbf{j} + 0.318\mathbf{k}) \\ &= -1060\mathbf{i} + 2120\mathbf{j} + 795\mathbf{k}\end{aligned}$$

$$F_x = -1060 \text{ N}$$

$$F_y = 2120 \text{ N}$$

$$F_z = 795 \text{ N}$$

Rectangular Components in Space



Solution

Noting that the components of the unit vector are the direction cosines for the vector, calculate the corresponding angles.

$$\begin{aligned}\lambda &= \cos\theta_x \mathbf{i} + \cos\theta_y \mathbf{j} + \cos\theta_z \mathbf{k} \\ &= -0.424\mathbf{i} + 0.848\mathbf{j} + 0.318\mathbf{k}\end{aligned}$$

$$\theta_x = 115.1^\circ$$

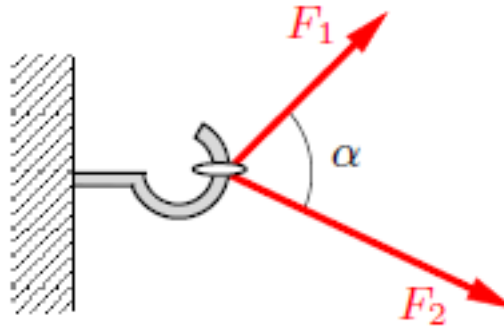
$$\theta_y = 32.0^\circ$$

$$\theta_z = 71.5^\circ$$

End

Problem

- Ex-2 A hook carries two forces F_1 and F_2 , as shown in figure, Determine the magnitude and direction of the resultant.

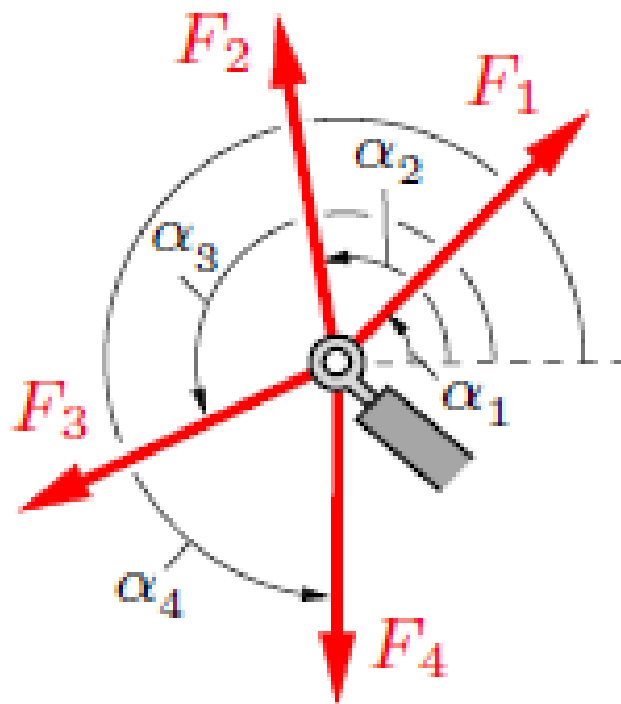


Problem

□ **Ex-3:** An eyebolt is subjected to four forces ($F_1 = 12 \text{ kN}$, $F_2 = 8 \text{ kN}$, $F_3 = 18 \text{ kN}$, $F_4 = 4 \text{ kN}$) that act under given angles

$$(\alpha_1 = 45^\circ, \alpha_2 = 100^\circ, \alpha_3 = 205^\circ, \alpha_4 = 270^\circ)$$

- ✓ with respect to the horizontal as shown in figure. Determine the magnitude and direction of the resultant.



Components of Force

Vector approach: Scalar components of F_3 can be obtained by writing F_3 as a magnitude times a unit vector $\mathbf{n}_{AB}$ in the direction of the line segment AB.

$$\mathbf{F}_3 = F_3 \mathbf{n}_{AB} = F_3 \frac{\overrightarrow{AB}}{AB} = 800 \left[\frac{0.2\mathbf{i} - 0.4\mathbf{j}}{\sqrt{(0.2)^2 + (-0.4)^2}} \right]$$

$$= 800[0.447\mathbf{i} - 0.894\mathbf{j}]$$

$$= 358\mathbf{i} - 716\mathbf{j} \text{ N}$$

$$F_{3_x} = 358 \text{ N}$$

$$F_{3_y} = -716 \text{ N}$$